



RippleTwin

Know the ripple before you close the road.

[Team name]

FEIT Hackathon 2026 · Challenge 5 · RPM Hire

Adding “USE RUSSELL ST” to one VMS board cut Lonsdale St delay by 45%* 10,493 → 5,746 vehicle-minutes

PROBLEM STATEMENT

- Councils and contractors struggle to foresee how a road closure — alone or **overlapping other works** — hits traffic, trams and pedestrians.
- Sign effects are guessed: in London, real diversions were **one-fifth** of what drivers said they would do (Chatterjee 2002).

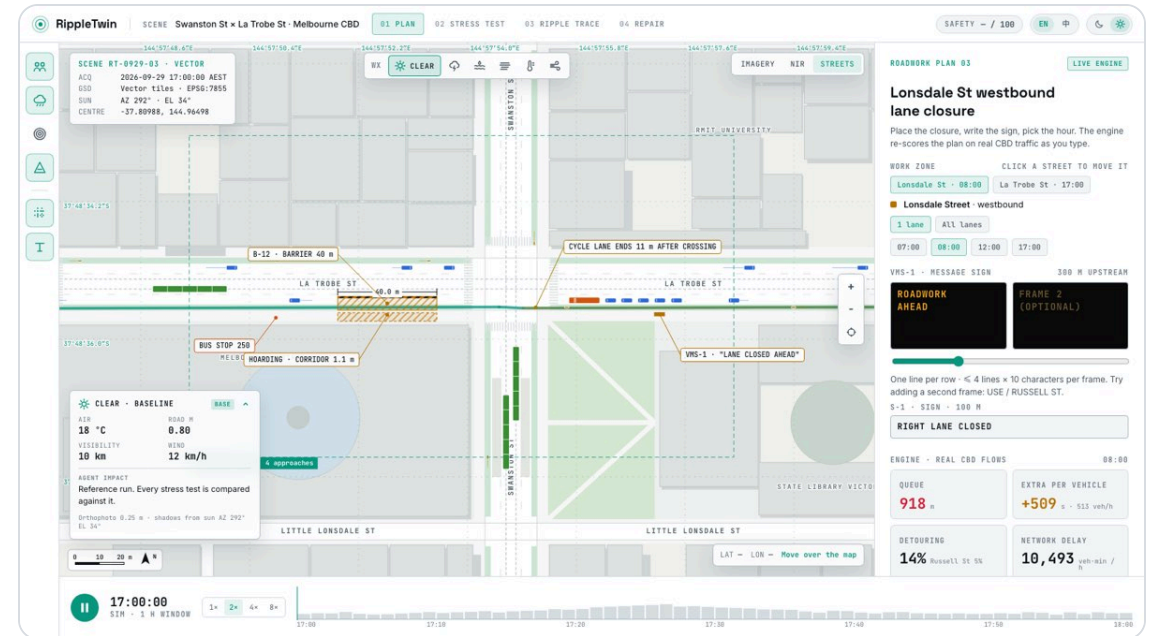
SOLUTION OVERVIEW

- 1 Close a lane on the **real Melbourne CBD** network; add barriers, signs and the exact **VMS wording**.
- 2 Four driver types — commuter, local, tourist, delivery — **read each sign** (keyword rules today, **AI** LLM next).
- 3 A deterministic engine computes **detours, queues and delay**, then re-tests each suggested fix.

AI interprets — the engine calculates.

WHO USES IT & PAYS

- Contractors & councils: impact evidence for each permit.
- RPM Hire: the value of each extra VMS or barrier → the hire quote (planned).



Live demo, 29 Sep: Lonsdale St lane closure, 8 am — engine on real CBD traffic counts

WHAT'S DIFFERENT

- RPM's VMS Preview shows how a sign **looks**; we show its **effect**.
- Mooven forecasts from past delays; we test the **wording** before deployment.
- one.network flags clashes; we count the **minutes they add**.

* Single run (1 lane closed, 8 am); keyword-rule sign reading; assumed traveller mix.

Tech & Progress

[Team name]

Real data, a working engine and a live demo — built since 29 Sep, 09:30

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TECH STACK

DATA

Python + OSMnx → static JSON, all from open data

ENGINE

JavaScript in the browser + Canvas map; deterministic, under 10 ms per scenario

SIGN READING

Keyword rules today; LLM next, with cached answers

HOSTING

Cloudflare Workers, one URL; API keys stay server-side

OPEN DATA

OpenStreetMap · DataVic SCATS & PTV GTFS · City of Melbourne

BUILT SO FAR

Live

public demo on Cloudflare since 29 Sep

1,513

CBD road links with 8 weeks of traffic-signal counts (SCATS)

–45%

Lonsdale St delay when the VMS adds “USE RUSSELL ST” (1 lane, 8 am)*

29,850

extra veh-min when two works overlap (3,515 apart; test grid) — a 5-day stagger → 0*

4 modes

cars, bikes, trams and pedestrians simulated at Swanston × La Trobe

16

equipment types from RPM Hire's range: barriers, signs, VMS

Try it: [demo link + QR](#) — team to confirm]

MVP BY 1 OCT

30 SEP

Clash cost on real CBD streets, shown in the app

1 OCT

Delay for tram riders and pedestrians (47 routes and 71 counters already loaded)

1 OCT

Advisor's 3 fixes — reword, move, stagger — compared on the CBD map

STRETCH

LLM sign reading with cached answers and ranges

1 OCT 10:00

Code freeze, tagged release, public repo

* Keyword-rule sign reading, single runs; 29 of 37 behaviour parameters (incl. the traveller mix) are assumed or low-confidence and flagged in the app; 74% of link flows are estimated from nearby counts. Data © OpenStreetMap contributors (ODbL); DataVic SCATS & GTFS (CC BY 4.0); City of Melbourne open data (CC BY). Status: 29 Sep 2026, 17:40 AEST.

RippleTwin — System Architecture

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